

(PRD) Inventory Single Source of Truth for SUR

Project Overview

SUR-SSOT (Same Unit Repair – Single Source of Truth) extends Asurion's DAX platform inventory SSOT—already proven for DSE/NDES with significant shrink reductions—to the Same Unit Repair business.

The project delivers a single API for SUR inventory availability and reservations, migrates inventory feeds and management from ServiceBench to DAX IVS/D365 where applicable, and eliminates UBIF → ServiceBench data replication in favor of a real-time orchestration approach.

OKR alignment: This project is under **Fulfillment Services 2026 Annual Plan, Objective #2: Create a unified fulfillment ecosystem with a single, accurate system of record for inventory.**

▼ Objectives

1. **Single API for SUR inventory** — Provide one endpoint for all SUR inventory availability inquiries and reservations, following the DES/ISP pattern (reservation ID in orders to UBIF and DAX).
2. **Migrate 3rd Party Non-Asurion-Owned Parts (NAOP) to DAX IVS** — Move inventory feeds from ServiceBench to DAX IVS for client-specific 3rd party repair networks (e.g. TELUS/Mobile Klinik, Mexico); support peril/BOM/equipment-type and provider-specific reservation strategy.
3. **Migrate 3rd Party Asurion-Owned (AOP) inventory to DAX** — Move inventory management from ServiceBench to DAX, exposed via IVS; Prism replaces ServiceBench for Asurion programs; inventory ownership in DAX, not Prism.
4. **Migrate UBIF NextGen Portal inventory to DAX IVS** — Replace UBIF → ServiceBench data replication with a global orchestration API that calls UBIF Current State and Next Gen in real time for part availability and quantity; include Distro Part Availability (Distro Vendor indicator) in the same API; deprecate inventory and distro feeds from UBIF to ServiceBench.
5. **Part price and claims** — Set partner invoice price transactionally on Job/Claim in ServiceBench so ServiceBench stops using replicated master price from UBIF; both UBIF Current State and Next Gen to support this; enable deprecation of price replication from UBIF. *(ServiceBench may retain local price copy for 3rd party claims; scope of that flow TBD.)*
6. **Eliminate replication risks for UBIF** — Address data inconsistency, latency, and scalability risks of replicating transactional part availability/quantity and master distro/price data from UBIF to ServiceBench by moving to real-time orchestration and transactional part price on claim.
7. **BAU Portal** — Timing and scope of BAU Portal feed migration are **TBD**; never assume in scope without explicit confirmation.

▼ Key Results

Related Objective #2 KR: 100% UBIF store migration to Next Gen Portal by end of Q2 2026; NextGen Portal MVP Feature Build. SUR-SSOT's UBIF orchestration and D365 migration depend on and align with this store migration timeline.

Key result	Team	Base	Q1 (<i>at risk</i>)	Q2	Q3	Q4
Migrate SUR inventory to	F&R, FF&C, IS, F&FS	SUR inventory currently	Migrate UBIF inventory	Migrate 3rd party repair	Migrate 3rd party Asurion	—

Key result	Team	Base	Q1 (<i>at risk</i>)	Q2	Q3	Q4
D365 (SSOT)		managed in ServiceBench	inquiries to D365	Non-Asurion owned parts to D365	owned parts inventory management to D365	
Viable plan to migrate Distro (SSOT)	F&R, FF&C, IS	—	—	—	—	Approved Distro Migration Plan
Reduce enterprise inventory shrink (<i>SUR contributes</i>)	F&R, FF&C, IS, F&FS	\$2.32M	—	—	\$250k Savings YTD YoY	\$600k Savings YTD YoY

Product Description

▼ Problem statement

SUR (Same Unit Repair) inventory is not served by a single source of truth. Today, inventory availability, JIT scheduling, and claims pricing depend on **data replicated from UBIF (Current State and Next Gen) into ServiceBench**. That replication creates material risks:

- **Data can become inconsistent** across UBIF and ServiceBench, so availability or price in one system may not match the other.
- **Latency** means ServiceBench may act on outdated inventory, so stores can fail to honor customer appointments or offer the wrong repair vs. replacement options.
- **Transactional data** (part availability and quantity by store) is high-frequency and highest-risk: inaccuracy leads to stores unable to honor appointments, replacement at repair cost when parts are under-stated, or missed revenue when parts are incorrectly shown as unavailable.
- **Master data** (distro part catalog, part price) supports JIT and financial reconciliation; when wrong or missing, JIT scheduling and Asurion–UBIF Franchise Co. reconciliation are wrong.

There is **no single API** for SUR inventory availability and reservations across repair channels. Third-party (NAOP and AOP) inventory is managed or fed through ServiceBench, and the enterprise remains **dependent on ServiceBench** for SUR inventory despite a proven DAX SSOT pattern for DSE/NDES that delivered significant shrink reductions. Extending that pattern to SUR will reduce shrink and labor cost and unify availability and reservations under one API.

▼ Historical Background

This problem was identified and evaluated before when SSOT was built for DES jobs. At the time, shrink had grown to millions of dollars annually, prompting the operations team to implement tight manual tracking processes — daily reporting, exception management, and follow-ups on missing parts.

It worked, and shrink dropped from **millions of dollars annually** down to **approximately \$100K** across the entire business. With shrink that low, the ROI on a system re-architecture couldn't compete with other priorities, and the project was shelved.

There was also a dependency and structural reason to wait: **NextGen Portal**, which would need to integrate with any consolidated inventory system, was still in early ideation. Building on a platform that was about to be replaced didn't make sense. Deprioritizing was the right call.

▼ Current state

- **UBIF → ServiceBench:** Three datasets are replicated from UBIF into ServiceBench:

1. **Part availability and quantity by store** (transactional) — Current State and Next Gen UBIF each send to ServiceBench via different integration patterns (Current State: events and APIs; Next Gen: events only). ServiceBench uses this locally for SUR Availability and JIT.
 2. **Distro part catalog** (master) — Indicates which parts are available for JIT from the UBIF Distribution Center. ServiceBench uses it for JIT scheduling.
 3. **Part price** (master) — Used in ServiceBench for claims and financial reconciliation between Asurion and UBIF Franchise Co.
- **ServiceBench** is the consumer of this replicated data for SUR Availability, Just in Time scheduling, and Claims. It also holds or receives **3rd party SUR inventory** (NAOP/AOP) via other mechanisms.
 - **DAX** already provides an inventory SSOT for DSE/NDES with proven shrink outcomes but does **not** yet serve SUR with a single availability/reservation API. UBIF store-level availability is owned by UBIF (Current State has an API; Next Gen F&O is system of record for migrated stores).
 - **Prism** is replacing ServiceBench for Asurion programs; inventory management for SUR is intended to move to **DAX** (exposed via IVS), not to Prism.

▼ Proposed solution

1. **Single API for SUR inventory** — One endpoint for all SUR inventory availability inquiries and reservations (DES/ISP pattern), with reservation ID in orders to UBIF and DAX. Consumers (e.g. ServiceBench, portals) call this API instead of relying on replicated UBIF data.
2. **Global orchestration API for UBIF** — Replace UBIF → ServiceBench replication for part availability and quantity with a **real-time orchestration API** that calls UBIF Current State and UBIF Next Gen. The same API returns a **Distro Vendor indicator** (Distro Part Availability), sourced from UBIF Current State until Next Gen has complete master inventory. Deprecate inventory and distro feeds from UBIF to ServiceBench.
3. **Part price on Job/Claim** — Set partner invoice price **transactionally on the Claim** in ServiceBench so ServiceBench stops using replicated master price from UBIF. Both UBIF Current State and Next Gen support this; price replication from UBIF is deprecated for UBIF claims.
4. **Migrate 3rd party to DAX IVS** — Move **3rd party Non-Asurion-Owned Parts (NAOP)** inventory feeds (e.g. TELUS/Mobile Klinik, Mexico) and **3rd party Asurion-Owned (AOP)** inventory management from ServiceBench to DAX, exposed via IVS. Phased rollout: UBIF/NAOP first, then 3rd party NAOP, then 3rd party AOP; pilot by location/provider with rollback approach for non-AOP.
5. **Single system of record** — D365/IVS and DAX become the inventory SOR for SUR where migrated; ServiceBench dependency for SUR inventory management is eliminated for these capabilities. Data integrity, reservations (including soft reservation and expiration where needed), and reconciliation are designed into the new flow.

Impacted Teams

Audience	Impact
UBIF (stores and distro)	Store and distro inventory exposed via orchestration API instead of replication; part price sent on claim; deprecation of outbound feeds to ServiceBench.
Service Bench / Prism	Service Bench: Becomes consumer of orchestration API for UBIF availability and distro; sets part price on Job/Claim; continues to handle 3rd party SUR inventory until migrated; eventual reduction in replicated UBIF data and related code. Prism: Replaces ServiceBench for Asurion programs; inventory management for SUR lives in DAX (IVS), not Prism; third-party AOP UI path (DAX MFE vs WMA) affects UX and rollout.
DAX / Engineering (F&R, FF&C, IS,	Build and operate orchestration API, IVS/D365 integration, and migration of NAOP/AOP feeds; own inventory SOR for SUR where in scope.

Audience	Impact
F&FS)	
3rd party repair (NAOP/AOP)	NAOP: TELUS/Mobile Klinik, Mexico, and other client-specific providers—feeds and reservation strategy move to DAX. AOP: Asurion-owned parts at 3rd party locations—inventory management moves to DAX; identity/Elite and on-demand ordering may change.
Store Ops / Remote Tech	SUR availability and JIT behavior change with single API and real-time data; store migration to Next Gen Portal and JIT/config (e.g. quantity-one, job type exclusions) affect operations.
Claims / Finance	Part price on claim affects reconciliation between Asurion and UBIF Franchise Co.; 3rd party claims may still use local price copy in ServiceBench (scope TBD).
Other Stakeholders	Scope, phasing, and timeline decisions; dependency on ServiceBench decommission, NextGen store migration, EA alignment, MDM team operations

Functional Requirements

Functional Requirements

Process Flows

Parts Management Flows

Out of Scope



Confirmed items that are **out of scope** for SUR-SSOT. Do not assume these are in scope unless explicitly confirmed in a future decision. Add new rows only when stakeholders explicitly confirm an item is out of scope.

Item	Reason Out of Scope	Notes
BAU Portal feed migration	Timing and scope are TBD; never assume in scope without explicit confirmation.	Do not assume in any quarter. If later confirmed in scope, add to roadmap and requirements.
Parts Portal	Not in scope for this SUR project.	Explicitly called out as out of scope in meeting with Prism team.
Microsoft IVS for UBIF inventory	Not recommended for this scope. Confluence: "Do not use Microsoft IVS for UBIF inventory for this scope."	Another replication approach; significant cost/effort; 3rd party SUR remains in ServiceBench either way.
Improve-existing-replication-only approach	Replication risks (especially for transactional data) are not addressed; approach is not in scope.	Architecture recommends orchestration API, not "improve replication only."